Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Network Topologies

Types of network topologies: Bus, Star, Ring, Mesh. Simulate a star topology using Cisco Packet Tracer.

Task: Draw a diagram of topologies with examples.

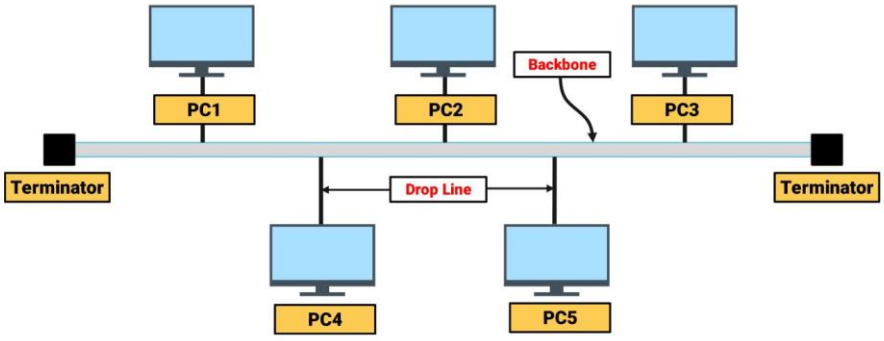
❖ **1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.**

➢ **Aim**

- To draw a diagram of a bus topology, label its key components, and explain one real-world example where this topology can be used.

➢ **Introduction**

- Bus topology is a type of network topology in which all computers and devices are connected to a single main cable called the **backbone cable**. Data travels along this cable and is available to the connected devices. Terminators are placed at both ends of the backbone cable to prevent signal reflection.



(Labelled Diagram of Bus Topology)

- **Key Components of Bus Topology**
  1. **Backbone Cable:** The main cable that connects all devices and carries network data.
  2. **Computers (End Devices):** Devices connected to the backbone cable to send and receive data.
  3. **Taps or Connectors:** Components used to connect each computer to the backbone cable.
  4. **Terminators:** Devices placed at both ends of the cable to prevent signals from reflecting back along the cable.

- **Working of Bus Topology**
- In bus topology, when a computer sends data, the data travels along the backbone cable. All connected devices can receive the signal, but only the

intended destination processes the data. Terminators absorb the signal at the ends of the cable and help prevent signal reflection.

- **Real-World Example**
  - A bus topology could be used in a small computer laboratory or a temporary network where a few computers need to communicate over a short distance. It requires less cabling and can be inexpensive to set up. However, it is rarely used in modern Ethernet LANs because a fault in the main backbone cable can disrupt the entire network.

- **Advantages**
  - Requires less cable than some other topologies.
  - Simple and inexpensive for small networks.
  - Suitable for small or temporary network setups.

- **Disadvantages**
  - A backbone cable failure can affect the entire network.
  - Troubleshooting cable faults can be difficult.
  - Performance may decrease when many devices share the same cable.

❖ **2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.**

➢ **Aim**

- To create a simple star topology in Cisco Packet Tracer using one switch and four PCs, with each PC named after a popular application.

➢ **Introduction**

- Star topology is a network topology in which all computers and other devices are connected to a central networking device, usually a switch. The switch manages communication between the connected devices. If one PC or its cable fails, the other PCs can continue working normally, provided the switch is functioning properly.

➢ **Requirements**

1. Cisco Packet Tracer software
2. One switch (Cisco 2960)
3. Four PCs
4. Copper Straight-Through cables

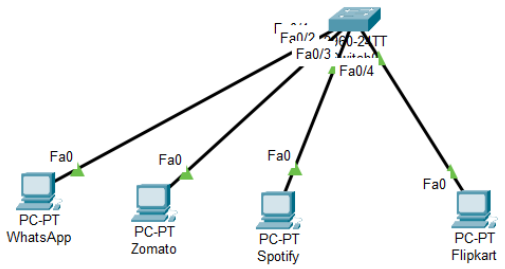
➢ **Procedure**

1. Open Cisco Packet Tracer.
2. Drag one Cisco 2960 switch and four PCs into the workspace.
3. Rename the PCs as follows:

- PC0 → WhatsApp

- PC1 → Zomato
- PC2 → Spotify
- PC3 → Flipkart

4. Connect each PC to a separate port on the switch using Copper Straight-Through cables.
5. For example, connect the PCs to switch ports FastEthernet0/1, FastEthernet0/2, FastEthernet0/3, and FastEthernet0/4.
6. Wait until the link indicators turn green.
7. Arrange the devices so that the switch is in the centre and all four PCs are around it.
8. Take a screenshot of the completed topology and insert it into the assignment.



(Star Topology Created in Cisco Packet Tracer Using One Switch and Four PCs.)

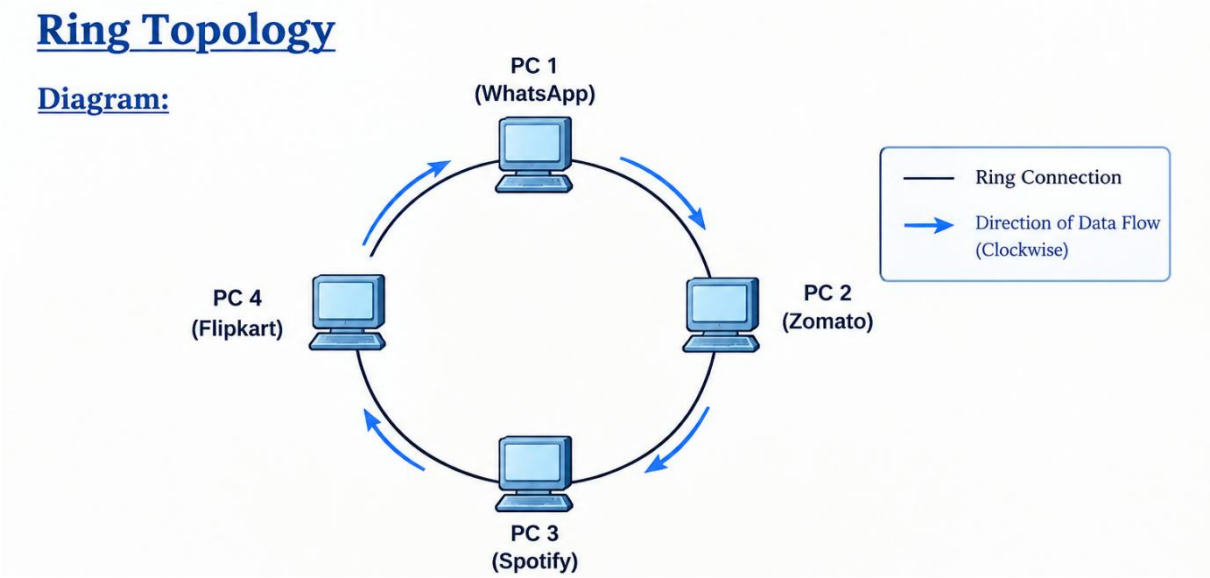
- **Working**
- In this star topology, all four PCs are connected to a central switch. When a PC sends data to another PC, the switch forwards the data to the appropriate destination port. Each PC has its own connection to the switch, making the network easy to manage and troubleshoot.

- **Advantages**
  - Easy to install and manage.
  - Failure of one PC or its cable does not normally affect the other PCs.
  - Faults can be identified easily.
  - Additional devices can be connected by using available switch ports.

- **Result**
  - A star topology was created successfully in Cisco Packet Tracer using one switch and four PCs named WhatsApp, Zomato, Spotify, and Flipkart.

❖ **3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup. Use arrows to show the direction of data flow.**

➢ **Ring Topology**



➢ **Explanation:**

- In a ring topology, data travels in a circular path from one device to the next in the direction shown by the arrows. Each device receives and checks the data, then forwards it to the next device until it reaches the intended destination.

❖ 4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

➢ Introduction

- Mesh and star are two types of network topologies. In a star topology, all devices are connected to a central switch or hub. In a mesh topology, devices are connected through multiple links, providing alternative paths for data transmission.

➢ Comparison Table

Topology	Advantages (Any Two)	Disadvantages (Any Two)
Star Topology	1. Easy to install, manage, and troubleshoot. 2. Failure of one device or its cable usually does not affect the other devices.	1. If the central switch or hub fails, the entire network may stop working. 2. It requires more cables than some simpler topologies, increasing installation costs.
Mesh Topology	1. Highly reliable because multiple connections can provide alternative paths for data. 2. Failure of one link usually does not interrupt communication if another suitable path is available.	1. Requires many cables and network ports, making it expensive. 2. Installation, configuration, and maintenance can be complex, especially in a full mesh network.